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Major Project Review–1 Evaluation Parameters

Parameter	Marks
Literature Survey	5
Base Paper Explanation	5
Technical Architecture	5
Dataset Strategy	5
Methodology & Tools	10
Total	30 Marks

Literature Survey (5 Marks)

Objective: To assess research depth and understanding of existing work.

What to Evaluate:

- Minimum **2–3 recent research papers** (preferably last 3 years)
- Papers from reputed sources: Scopus and SCI Indexed Journal Papers
 - IEEE Transactions
 - Springer
 - Elsevier
 - arXiv (for AI domains)
- Clear understanding of:
 - Existing models
 - Techniques used
 - Limitations in prior work
- Identification of **research gap**
- Comparison table of related works

Expected from Students:

- Structured summary of each paper
- Comparative analysis table
- Clear explanation of how their project differs/improves

Marks Distribution (5)

Criteria	Marks
Number & Quality of Papers	1
Understanding & Explanation	1
Research Gap Identification	2
Comparative Analysis	1

Base Paper Explanation (5 Marks)

Objective: To assess understanding of the primary reference paper guiding the project.

What to Evaluate:

- Clear explanation of:
 - Problem addressed in base paper
 - Model architecture
 - Algorithms used
 - Evaluation metrics
- Mathematical understanding
- Limitations of base paper
- Scope for improvement
- Model flow
- Loss functions
- Optimization method
- Dataset used in base paper

Marks Distribution (5)

Criteria	Marks
Problem & Objective Clarity	1
Architecture Explanation	2
Technical Depth	1
Limitations & Improvements	1

Technical Architecture Design of Proposed System (5 Marks)

Objective: To evaluate system design clarity and feasibility.

What to Evaluate:

- Proper architecture diagram
- Module breakdown
- Data flow representation
- Justification for model selection
- Use of LLM (Open-source or API-based)
- RAG architecture (if used)
- Prompt pipeline
- Embedding model
- Vector database (if applicable)
- Agent workflow
- Tool integration
- Planning & reasoning strategy
- Memory module

Marks Distribution (5)

Criteria	Marks
Architecture Clarity	2
Technical Justification	1
System Modules	1
Feasibility & Scalability	1

Dataset Strategy (5 Marks)

Objective: To assess planning for data collection and preparation.

What to Evaluate:

- Dataset source (Kaggle / UCI / Custom / Synthetic)
- Data size adequacy
- Data preprocessing strategy
- Train-test-validation split
- Ethical & bias considerations
- Corpus source
- Data cleaning process
- Chunking strategy (if RAG)
- Image/Text/Tabular data handling
- Augmentation strategy

Marks Distribution (5)

Criteria	Marks
Dataset Relevance	2
Preprocessing Plan	2
Ethical & Bias Consideration	1

Methodology & Tools (10 Marks)

Objective: To evaluate technical approach and technology stack selection.

What to Evaluate:

- Framework selection (PyTorch / TensorFlow / Hugging Face / LangChain)
- Training strategy
- Hardware planning (GPU usage)
- Deployment tools (Flask, Streamlit, FastAPI, etc.)
- Version control (GitHub)

Students Should Explain:

- Why this framework?
- Why this model?
- Why this optimization technique?

Marks Distribution (10)

Criteria	Marks
Tool Selection Justification	4
Implementation Strategy	4
Deployment Plan	2